



IoT-Based Smart Home Automation System

Transforming residential living through intelligent connectivity and automation

Learn more about IoT

What is Smart Home Automation?



Smart home automation leverages Internet of Things (IoT) technology to create an interconnected ecosystem where household appliances and systems communicate seamlessly. Devices ranging from lighting and climate control to security systems can be managed from a unified platform.

The primary goal: deliver enhanced convenience, improved safety protocols, and significant energy efficiency through intelligent automation rather than manual operation.

Problem Statement

Home appliances are often left on unnecessarily, leading to avoidable energy waste and higher costs. A smart IoT system is needed to automate appliance usage, improve efficiency, and ensure energy is consumed only when needed.



Challenges of Manual Appliance Control



Energy Wastage from Oversight

Manual control often means forgotten appliances running idle. Average households waste 10-15% energy yearly, equating to £160-£200 in losses. Lights left on overnight contribute significantly, driven by inconsistent habits and lack of awareness.



Inconvenience in Daily Routines

Users experience delays and missed opportunities for optimisation, such as adjusting heating or cooling remotely. Surveys indicate 60% frustration from manual checks, disrupting schedules and potentially reducing productivity by 1-2 hours weekly.



Lack of Real-Time Insights

Without visibility into usage patterns, informed decisions on energy optimisation are impossible. Peak-hour overuse can spike costs by 20%. For example, one family's overlooked fan runtime increased their bill by 18%.

The Energy Wastage Challenge:

Manual Control Limitations

Traditional switches require physical presence and constant attention, leading to forgotten devices left running unnecessarily.

Unnecessary Consumption

Lights, fans, and appliances often remain powered when no one is in the room, wasting valuable energy resources.

Lack of Monitoring

Without real-time visibility, homeowners remain unaware of which devices are consuming power and when.

A smart IoT system addresses these challenges by automating device management and optimizing energy usage patterns based on actual need.



Drawbacks of Existing Systems

High Hardware Costs

Systems depend on sensors, switches, and other devices that increase upfront expenses.

Installation Complexity

Setup often requires wiring changes and professional installation.

Limited Flexibility

Existing systems are difficult to modify, expand, or adapt to new requirements.

Limited Testing Capabilities

Automation cannot easily be tested without access to physical devices.

User Difficulty

Some systems are hard for beginners to understand and use effectively.

Ongoing Maintenance

Hardware upkeep and repairs add recurring effort and cost over time.

How Our System Is Different

Simple and Flexible Platform

Designed for home automation with an easy, adaptable setup.

Easy Testing Without Hardware

Can be tested and demonstrated without a full physical device setup.

Low Cost and Beginner-Friendly

Affordable to use and easy for new users to understand and operate.

Quick Device Management

Devices can be added or removed quickly as needs change.

Clean Monitoring Dashboard

Provides real-time monitoring through a clear and simple dashboard.

Energy Reduction Focus

Automation rules help reduce unnecessary energy use.



Our IoT-Based Solution

01

Centralized Control Hub

Single mobile app or web dashboard to manage all connected devices

02

Intelligent Automation

Programmable rules automatically switch devices ON/OFF based on schedules or conditions

03

Real-Time Monitoring

Live status updates and energy consumption tracking for all appliances

Key Objectives



Reduce Energy Wastage

Eliminate unnecessary power consumption through automated device management and smart scheduling.



Improve Safety

Enhanced security through remote monitoring and automated responses to unusual activity.



Remote Access

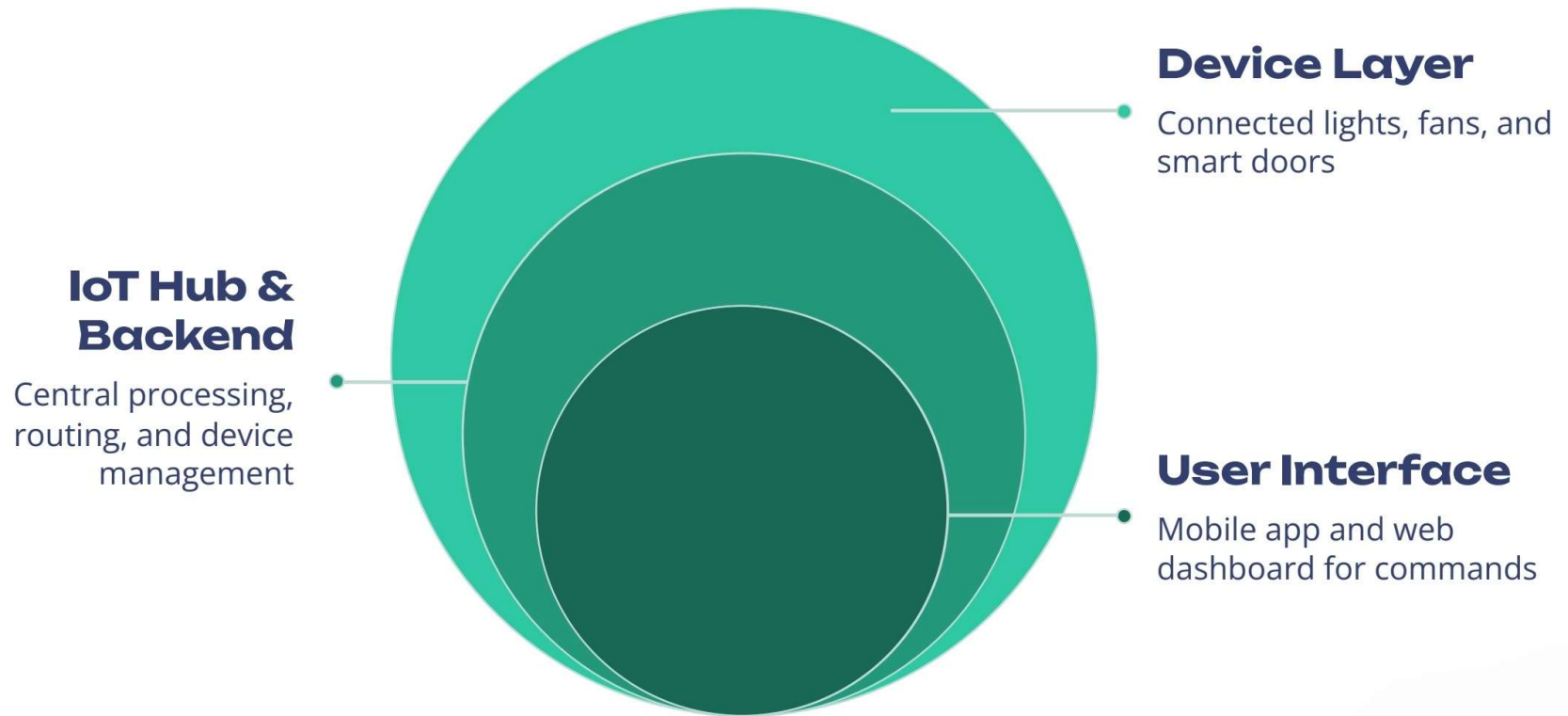
Control home appliances from anywhere using internet-connected devices.



Enable Automation

Set up rules and schedules to automate routine tasks without manual intervention.

System Architecture



How It Works



User Command

Owner sends control instruction via mobile app



System Processing

Backend interprets request and validates authorization



Device Update

Target appliance receives ON/OFF command instantly



Status Feedback

Real-time confirmation displayed to user interface

Automation rules execute independently based on predefined conditions such as time, sensor input, or occupancy detection.

Learn more about us

Core Features

Remote ON/OFF Control

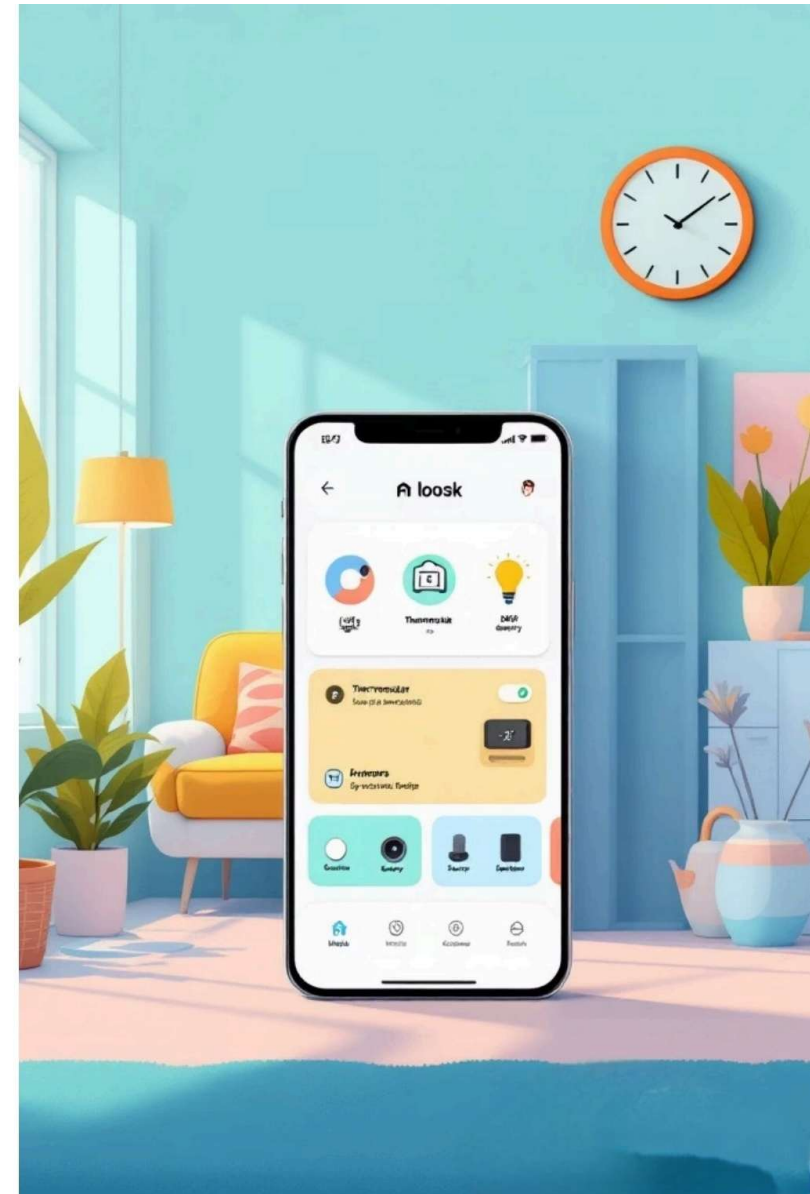
Switch devices from anywhere using internet-connected smartphone or tablet

Real-Time Status

Instant visibility of which devices are active and their current state

Expandable Design

Start with basic devices and add more appliances as needed





Expected Outcomes

Reduced Energy Consumption

Automated control eliminates wasted power from forgotten devices, leading to lower utility bills.

Convenient Home Management

Control everything from one place instead of searching for multiple switches and remotes.

Enhanced Monitoring

Real-time visibility into appliance status provides peace of mind and security awareness.

Future Enhancements

1

Voice Assistant Integration

Control devices using natural voice commands through Alexa, Google Assistant, or Siri

2

AI-Powered Predictions

Machine learning algorithms learn patterns and suggest optimal automation rules

3

Detailed Energy Reports

Comprehensive analytics showing consumption patterns and cost savings over time

4

Hardware Integration

Connect to actual smart devices and sensors for real-world deployment

Thank You